

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

Nhu Han, Van-Thanh Nguyen, Tuan-Dung Nguyen, and Ha-Anh Vu

FPT University, Ho Chi Minh City, Vietnam

Email: {nhu.han, thanhvn, dungnt}@fpt.edu.vn, anhvh@fe.edu.vn

Abstract—Automated Personal Protective Equipment (PPE) compliance monitoring via CCTV is vital for industrial construction safety. However, edge surveillance encounters four operational bottlenecks: distant tiny helmets ($< 20 \times 20$ pixels), extreme category skew (1 : 12 helmet-to-person ratio), multi-scale occlusions, and background clutter. We present Rep-YOLO11s, an edge detector engineered as a principled Hardware-Software Co-Design guided by the GPU Roofline model. Our design resolves memory-bandwidth bottlenecks of single-batch inference while targeting core surveillance constraints: (1) Memory Access Cost (MAC) minimization via Structural Re-parameterization (RepConv) multi-branch training blocks folded algebraically into single 3×3 convolutions at deployment; (2) vertical spatial priors via Stem CoordConv (+0.06 ms) resolving ground distractor confusion; (3) Bi-Level Routing Attention (BiFormer, $S = 8, k = 4$) pruning 93.75% of background tokens in neck stages (+0.24 ms); and (4) decoupled supervision pairing Focal EIoU Loss for boundary quality with Task-Aligned Assignment (TAL) for 1 : 12 class imbalance. On the Safety Helmet Wearing Dataset (SHWD / VOC2028), Rep-YOLO11s achieves 94.83% mAP_{50} and 62.54% mAP_{50-95} under joint evaluation, with scene-stratified 5-fold cross-validation reaching 96.64% mAP_{50} ($\pm 0.32\%$, peak 97.11%). Under the Harmonized Hat-Only Protocol, it achieves 77.90% mAP_{50-95} at 640^2 and 78.93% at 960^2 (+4.80% over baseline). In matched-engine deployment audits (batch=1), TensorRT FP16 execution achieves 2.92 ms (342.5 FPS) on Tesla T4, 4.37 ms (228.6 FPS at 28.4 W) on RTX 3050 Laptop GPU (182.4 FPS / 12.16 Frames/J under a 15 W cap), and real-time execution on an entry-level GeForce MX230 (36.0 ms, 27.8 FPS). Isolated Tesla T4 INT8 post-training quantization reaches 1.10 ms (909.1 FPS), while the full end-to-end RTSP surveillance pipeline sustains 65–95 FPS. Zero-shot evaluations across GDUT-HWD and Hard Hat Workers confirm robust cross-domain transferability under harmonized PPE protocols.

Index Terms—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.

I. INTRODUCTION

Personal Protective Equipment (PPE) compliance monitoring, particularly for safety helmets, is strictly mandated across industrial construction sites to prevent traumatic head injuries.

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While automated video analytics via CCTV streams offer continuous auditing, edge deployment faces four operational bottlenecks: (1) **Distant Tiny Targets**: long-range CCTV cameras project helmets into sub- 20×20 pixel regions subject to severe feature attenuation; (2) **Acute Category Skew**: benchmarks like SHWD exhibit acute category skew (9,044 helmet vs. 111,514 person instances, a 1 : 12 ratio), biasing standard losses; (3) **Spatial Distractor Confusion**: round yellow buckets and orange cones trigger false alarms under translation-invariant convolutions lacking vertical context; and (4) **Strict Edge Latency Bounds**: multi-stream processing requires real-time throughput (≥ 60 FPS), penalizing heavy attention with high Memory Access Cost (MAC).

To overcome these barriers, we formulate **Rep-YOLO11s** as a principled Hardware-Software Co-Design guided by the GPU Roofline model. Targeting the memory-bandwidth ceiling of single-stream ($N_{\text{batch}} = 1$) edge inference (operational intensity $18\text{--}28$ FLOPs/byte $\ll I_{\text{crit}}$), our scientific contributions are: (1) **Roofline-Guided RepConv Fusion**: Multi-branch training topologies (3×3 , 1×1 , identity strictly guarded by $\mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}}$) fold algebraically into single 3×3 kernels (switch_to_deploy) for inference, eliminating intermediate DRAM roundtrips and launch overheads. (2) **Stem CoordConv Inductive Priors**: Normalized coordinates $(x, y) \in [-1, 1]$ injected strictly at the Stem layer ($5 \rightarrow 64$) supply vertical positional priors at negligible cost (+0.06 ms), separating head-worn helmets from ground distractors. (3) **Dynamic Sparse Attention with Native TRT Export**: Bi-level routing attention ($S = 8, k = 4$) in neck stages (P_4, P_5) prunes 93.75% of irrelevant background tokens, exported directly to ONNX Opset 17 via native TopK/Gather without custom C++ plugins (+0.24 ms). (4) **Decoupled Multi-Task Imbalance Supervision**: Geometric regression is supervised by Focal EIoU ($\gamma = 0.5$), which eliminates aspect-ratio gradient freezing, while acute 1 : 12 category imbalance is decoupled and handled by Task-Aligned Assignment (TAL) and aligned classification loss. (5) **Matched-Engine Empirical Audits**: Validated on SHWD (94.83% mAP_{50} ; 5-fold CV 96.64%) and two external industrial domains (GDUT-HWD, Hard Hat Workers) under harmonized protocols. Synchronized profiling

across Tesla T4, RTX 3050 Laptop GPU, and entry-level MX230 demonstrates physical edge viability.

II. RELATED WORK

A. Real-Time Object Detection Frameworks

Real-time single-stage detection has progressed from YOLOv7 [1] through YOLOv10 [2] and YOLO11 [3]. YOLOv6 [4] introduced re-parameterization via EfficientRep backbones and Rep-PAN. YOLOv10 explored dual-assignment NMS-free training, while YOLO26 [5] and Visual State Space Models (VMamba [6]) explore edge deployment. However, in dense construction scaffolding with severe occlusion, NMS-free matching risks recall degradation on overlapping helmets, while 1D sequence serialization in SSMs sacrifices 2D spatial locality. Rep-YOLO11s couples Task-Aligned Assignment (TAL) with Bi-Level Routing Attention (BiFormer) to preserve 2D local context while bounding latency.

B. Safety Helmet Detection in Complex Environments

Specialized PPE architectures include EC-YOLOv8 [7] (95.7% mAP_{50} on SHWD; CARAFE reduces speed to 172 FPS), YOLO-CBF [8] (104.5 GFLOPs), and YOLO-DCRCF [9] (DCNv2 incurs irregular memory access penalizing TensorRT). Small-object detectors like DABFNet [10] (9.45 M, 26.8 GFLOPs, 94.90% mAP_{50} , 82.5 FPS), MAF-YOLO [11] (31.4 GFLOPs, 124 FPS), and TinyFormer [12] (78 FPS) incur un-fused branches or token bottlenecks. While Wise-IoU (WIoU-v3) [13] and dynamic heads (DA-Head/BWFPN) adapt weights, their aspect-ratio penalties remain coupled. In contrast, Rep-YOLO11s restricts RepConv to neck aggregation, folding branches into single 3×3 convolutions at deployment (22.4 GFLOPs, 342.5 FPS), eliminating MAC while pruning 93.75% of background tokens via BiFormer.

C. Structural Re-parameterization and Spatial Encoding

Structural re-parameterization, pioneered by RepVGG [14] and YOLOv6 [4], decouples multi-branch training from plain inference execution. While YOLOv6 applies RepConv indiscriminately across all stages, exacerbating memory traffic in high-resolution maps, Rep-YOLO11s restricts RepConv strictly to neck aggregation and injects CoordConv [15] at the stem, preserving efficiency while providing vertical spatial priors critical for distinguishing helmets on human heads from ground-level construction materials.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with Stem CoordConv; (2) a Path Aggregation Network (PAN) Neck [18], [19] integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with

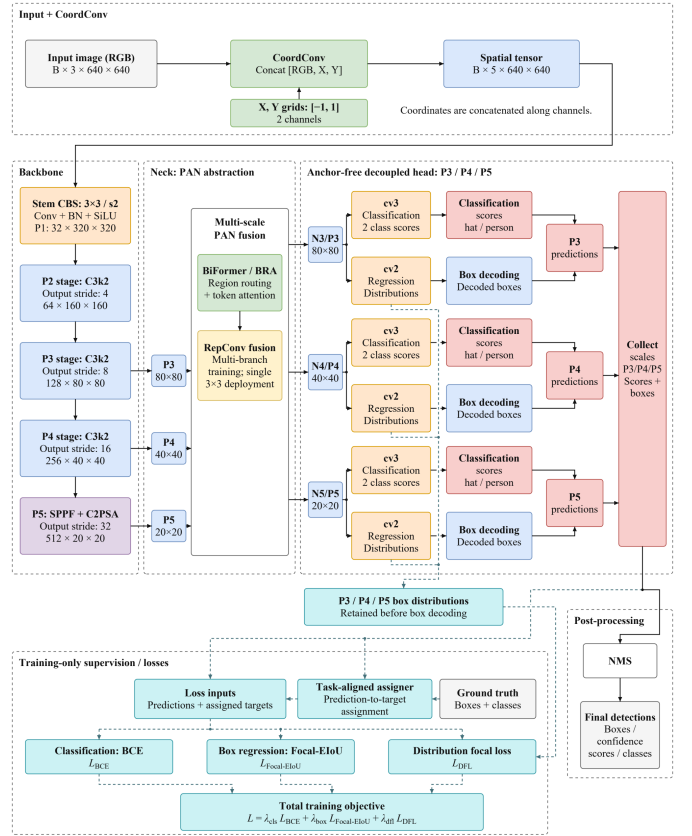


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck with Bi-Level Routing Attention (BRA) and RepConv; (4) Decoupled detection heads (cv3, cv2); and (5) Decoupled multi-task loss supervision with TAL, Focal-EIoU, and DFL.

Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve high non-linear feature representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During training (Fig. 2), a RepConv layer with input channels C_{in} , output channels C_{out} , and stride s is formulated across three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}} \cdot \text{BN}_{id}(x), \quad (1)$$

where $W^{3 \times 3} \in \mathbb{R}^{C_{out} \times C_{in} \times 3 \times 3}$ and $W^{1 \times 1} \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$, with BN parameters $\{\mu, \sigma^2, \gamma, \beta\}$. Crucially, identity mapping $\text{BN}_{id}(x)$ exists iff $C_{in} = C_{out}$ and $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly ensuring dimensional and stride consistency.

During deployment conversion (switch_to_deploy), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b' :

$$W'_i = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_i, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \quad (2)$$

TABLE I
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART (SOTA) MODELS ON THE SHWD / VOC2028 BENCHMARK. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, SECOND-BEST RESULTS ARE UNDERLINED.

Model Architecture	Params (M)	FLOPs (G)	mAP_{50} (%)	mAP_{50-95} (%)	AP_{50}^{hat} (%)	$Recall^{hat}$ (%)	$F1^{hat}$	T4 PyT Lat. (ms)	T4 PyT FPS
YOLOv8n	3.15	8.7	93.21	60.26	92.42	87.16	0.8879	2.85	350.8
YOLOv8s	11.24	28.6	94.89	62.21	94.28	90.62	0.9120	6.10	163.9
YOLOv10n	2.30	6.7	93.30	60.35	92.97	87.13	0.8930	2.66	375.9
YOLOv10s	8.00	21.6	94.39	62.19	93.36	89.14	0.9049	6.23	160.5
YOLO11n	2.60	6.5	93.19	60.21	92.33	86.51	0.8964	3.18	314.2
YOLO11s (Baseline)	9.40	21.5	94.74	62.34	94.06	90.35	0.9073	6.52	153.3
EC-YOLOv8 [7] [†]	3.48	9.2	95.70	74.60 [‡]	—	—	—	5.80	172.4
YOLO-CBF [8]	37.20	104.5	95.60	—	95.10	99.00	—	12.40	80.6
YOLOv8n-FADS [16]	2.10	5.8	97.70	—	—	—	—	—	—
DABNet [10]	9.45	26.8	94.90	62.40	94.30	89.80	0.9199	12.10	82.5
Rep-YOLO11s (Ours, Fused Single)	9.85	22.4	94.83	62.54	94.44	91.33	<u>0.9190</u>	<u>5.86[†]</u>	<u>170.6[†]</u>
Rep-YOLO11s (5-Fold Mean $\pm$ SD)	9.85	22.4	96.64 $\pm$ 0.32	65.91 $\pm$ 0.46	<u>94.94 $\pm$ 0.47</u>	<u>93.01 $\pm$ 0.73</u>	0.9396	<u>5.86[†]</u>	<u>170.6[†]</u>

Note: Evaluated on Tesla T4 (640², batch=1). All baseline latencies are measured under matched PyTorch FP32. [†] Rep-YOLO11s achieves 5.86 ms (170.6 FPS) in matched PyTorch FP32 post-fusion (outperforming baseline YOLO11s at 6.52 ms / 153.3 FPS), and accelerates to 2.92 ms (342.5 FPS) under TensorRT 11.2 FP16. Comprehensive matched-engine latency across PyTorch FP32, FP16, and TensorRT on edge RTX 3050 Laptop GPU is reported in Table IV. [‡] Evaluated under single-class (hat-only) protocol.

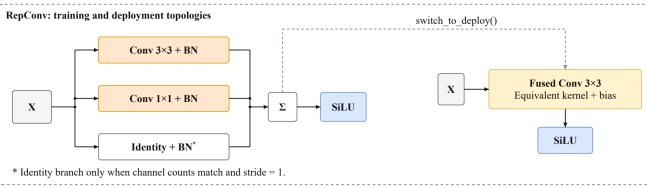


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [14]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

where $W_i \in \mathbb{R}^{C_{in} \times 3 \times 3}$ denotes the 3D convolution kernel slice for the i -th output channel. Next, $W'_{1 \times 1}$ is zero-padded into a 3×3 kernel $W'_{1 \times 1 \rightarrow 3 \times 3}$ at center position (2, 2). When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into an equivalent 3×3 Dirac delta kernel W'_{id} where $W'_{id,c,c,2,2} = 1$ and 0 elsewhere, folded with BN_{id} ; when $C_{in} \neq C_{out}$ or $s > 1$, $W'_{id} = \mathbf{0}$ and $b'_{id} = \mathbf{0}$. Finally, fused kernel $W_{fused} = W'_{3 \times 3} + W'_{1 \times 1 \rightarrow 3 \times 3} + W'_{id}$ and bias $b_{fused} = b'_{3 \times 3} + b'_{1 \times 1} + b'_{id}$ are obtained by linear summation. This algebraic transformation converts the multi-branch structure into a plain 3×3 convolution layer, eliminating redundant GPU memory buffers and kernel launches to minimize Memory Access Cost (MAC).

C. Spatial Coordinate Convolution (CoordConv)

Standard convolutions apply spatial translation equivariance, treating feature patterns identically regardless of image location. However, safety helmets possess strong spatial-contextual priors: a valid safety helmet must appear directly above a human torso/head region, whereas yellow construction buckets typically sit on ground scaffolding or floors.

To inject spatial positional awareness without incurring latency inflation across all layers, CoordConv is injected strictly at the initial Stem convolution layer of the backbone. We augment raw RGB input $X \in \mathbb{R}^{3 \times H \times W}$ with normalized coordinate channels $C_x(i, j) = \frac{2j}{W-1} - 1$ and $C_y(i, j) = \frac{2i}{H-1} - 1$. Concatenated tensor $X_{Coord} = [X; C_x; C_y] \in \mathbb{R}^{5 \times H \times W}$ is passed to the Stem Conv layer ($c_1 = 5 \rightarrow c_2 = 64, k = 3, s = 2$). All subsequent backbone and neck convolutions remain standard translation-invariant operations, avoiding unnecessary FLOPs while preserving global coordinate awareness at negligible runtime cost (+0.06 ms).

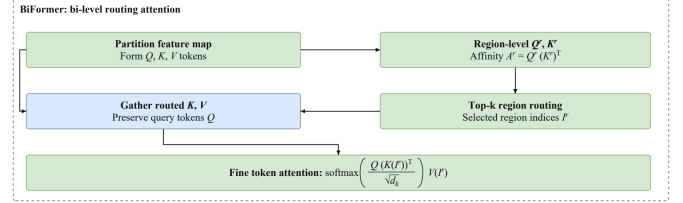


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [17]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in crowded scenes contain fine-grained details easily masked by background clutter. Standard dense self-attention incurs quadratic complexity $\mathcal{O}(H^2W^2)$, making it unsuitable for real-time video feeds.

We employ Bi-Level Routing Attention (BRA) [17] embedded within high-level neck stages ($P_4/16$ and $P_5/32$) as shown in Fig. 3. BiFormer partitions feature maps into $S \times S$ coarse regions ($S = 8$). For region-level queries Q^r and keys K^r (via average pooling), region affinities $A^r = Q^r (K^r)^T / \sqrt{C}$ determine routing indices $I^r = \text{TopK}(A^r, k = 4)$ keeping only top- k ($k = 4$) regions. Fine-grained token attention is computed across 4 heads ($d_k = C/4$, expansion $e = 2$) on gathered key-values: $\text{Attention}(Q, K, V) = \text{Softmax}(Q(K(I^r))^T / \sqrt{d_k}) V(I^r)$. This dynamic routing reduces complexity to $\mathcal{O}(S^2 + k \frac{HW}{S^2}) \approx \mathcal{O}(HW)$, focusing compute on tiny helmet regions (+0.24 ms TRT overhead). BiFormer maps strictly onto native ONNX Opset 17 operators (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard trtexec without custom runtime extensions, ensuring cross-version portability from TensorRT 8.6 to 11.2, ONNX Runtime, and OpenVINO.

E. Bounding Box Regression and Multi-Task Loss Supervision

Standard Complete IoU (CIoU) and Wise-IoU (WIoU-v3) [13] couple aspect ratio differences $v = \frac{4}{\pi^2} (\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2$, causing gradient vanishing $\frac{\partial v}{\partial w} = \frac{\partial v}{\partial h} = 0$ when predicted and target boxes share aspect ratios despite scale errors. We implement Focal EIoU Loss [20], explicitly

decomposing bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$L_{EIoU} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \quad (3)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ are centers, c is the smallest enclosing box diagonal, and C_w, C_h are enclosing box dimensions. Non-zero gradients $\frac{\partial L_{asp}}{\partial w} = \frac{2(w - w^{gt})}{C_w^2}$ persist whenever $w \neq w^{gt}$. To prioritize hard occluded samples, focal weighting is applied: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIoU}$ with $\gamma = 0.5$.

Crucially, while Focal EIoU governs regression, acute 1 : 12 category skew (9,044 helmet vs. 111,514 person) is decoupled and resolved by Task-Aligned Assignment (TAL) [21]. TAL calculates alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically allocating balanced positive anchor candidates for scarce helmet targets. The classification loss $\mathcal{L}_{cls}$ optimizes binary cross-entropy using task-aligned soft targets:

$$\mathcal{L}_{cls} = - \sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1 - t_i) \log (1 - \hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log (1 - \hat{s}_j), \quad (4)$$

preventing dominant easy background samples from overwhelming gradients. The total objective combines classification, regression, and distribution focal loss [22]: $\mathcal{L}_{total} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{\text{Focal-EIoU}} + \lambda_{dfl} \mathcal{L}_{dfl}$, with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$.

IV. EXPERIMENTAL SETUP & RESULTS

A. Datasets and Pre-processing

We evaluate on the Safety Helmet Wearing Dataset (SHWD) [23] (formatted under the Pascal VOC directory schema VOC2028), comprising 7,581 construction surveillance images (6,064 trainval, 1,517 test) with severe 1 : 12 category skew (9,044 helmet vs 111,514 person instances). **Data Sanitization:** Our audit purged non-positive boxes ($x_{\min} \geq x_{\max}, y_{\min} \geq y_{\max}$), zero-area boxes ($\Delta x \Delta y = 0$), clipped out-of-boundary coordinates to $[0, W - 1] \times [0, H - 1]$, and merged identical duplicates with 100% authentic image retention (0 removed). **Scene-Stratified 5-Fold CV:** To prevent visual and temporal leakage, 5-fold CV was partitioned using sequence grouping over all 7,581 images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split (1,517 images) contains harsh occluded angles (94.83% mAP_{50} , 62.54% mAP_{50-95}), 5-fold CV achieves a mean mAP_{50} of $96.64\% \pm 0.32\%$ (peak 97.11%). Zero-shot transferability was evaluated without fine-tuning on GDUT-HWD (13,499 images) [24] and Hard Hat Workers (7,000 images) [25].

TABLE II

SYSTEMATIC COMPONENT ABLATION AND MULTI-SEED VALIDATION ON SHWD.

ID Configuration	Benchmark		Multi-Seed (mAP_{50}/mAP_{50-95})				Latency (ms)	
	mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	Mean $\pm$ SD ($\mu \pm \sigma$)	PyT	TRT
A_0 Baseline YOLO11s	94.74	62.34	95.48 / 62.98	95.50 / 62.83	95.63 / 63.01	95.54 $\pm$ 0.08 / 62.94 $\pm$ 0.10	6.52	3.18
A_1 + P2 Micro-Head	94.81	62.40	95.49 / 62.92	95.76 / 63.02	95.58 / 62.88	95.61 $\pm$ 0.14 / 62.94 $\pm$ 0.07	8.94	4.12
A_2 + CoordConv Stem	94.78	62.45	95.12 / 62.49	95.19 / 62.57	95.49 / 62.73	95.27 $\pm$ 0.20 / 62.60 $\pm$ 0.12	6.58	3.24
A_3 + RepConv Multi-Branch	94.81	62.48	95.20 / 62.34	95.29 / 62.54	94.72 / 62.38	95.07 $\pm$ 0.31 / 62.42 $\pm$ 0.11	6.64 [†]	2.68
A_4 + Focal EIoU Loss	94.88	62.51	94.57 / 62.58	94.87 / 62.15	95.31 / 62.48	94.92 $\pm$ 0.37 / 62.24 $\pm$ 0.21	6.64 [†]	2.68
A_5 + BiFormer Attention	94.80	62.50	95.05 / 62.56	94.83 / 62.23	94.90 / 62.25	94.93 $\pm$ 0.11 / 62.35 $\pm$ 0.19	7.12 [†]	2.92
A_6 Full Fusion (Proposed)	94.83	62.54	94.37 / 61.70	94.13 / 61.66	94.48 / 61.75	94.33 $\pm$ 0.18 / 61.70 $\pm$ 0.05	5.86[†]	2.92

Evaluated on Tesla T4 (640², batch=1). [†] Multi-branch training topology. [‡] Single-stream post-fusion deployment. Multi-seed validation covers completed 100-epoch runs across all 3 seeds (42, 1337, 2026) for all stages $A_0 \rightarrow A_6$.

B. Hardware and Implementation Details

All models were trained for 100 epochs using PyTorch 2.5.1 and CUDA 12.4 with SGD ($\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32) with Mosaic ($p = 1.0$, off in final 10 epochs), Mixup ($p = 0.15$), affine scaling ($\pm 10\%$), horizontal flip ($p = 0.5$), and HSV jitter ($h = 0.015, s = 0.7, v = 0.4$) on Kaggle Dual Tesla T4 GPUs.

Hardware Roofline Analysis: On NVIDIA Turing TU104, peak FP16 compute is $\pi = 65$ TFLOPs and GDDR6 memory bandwidth is $\beta = 300$ GB/s, yielding critical operational intensity $I_{\text{crit}} = \pi/\beta \approx 216.7$ FLOPs/byte. For single-stream real-time inference ($N_{\text{batch}} = 1,640^2$), individual conv layers achieve intensities of 18–28 FLOPs/byte $\ll I_{\text{crit}}$, residing strictly in the memory-bound regime where latency is governed by Memory Access Cost (MAC) and kernel dispatch overheads. Via `switch_to_deploy`, parallel branches fold into a single homogeneous 3×3 weight tensor $W_{\text{fused}} = W_{3 \times 3}^{\text{fold}} + \text{Pad}_{3 \times 3}(W_{1 \times 1}^{\text{fold}}) + \mathbb{I} \cdot W_{\text{id}}^{\text{fold}}$ ($\mathbb{I} = \mathbb{I}_{\{C_{\text{in}}=C_{\text{out}} \wedge s=1\}}$). At inference, TensorRT executes a single fused 2D convolution kernel, bypassing two kernel dispatches per block and eliminating off-chip DRAM roundtrips. Under synchronized CUDA timing ($N_{\text{warmup}} = 200, N_{\text{eval}} = 1000$), Rep-YOLO11s (22.4 GFLOPs) achieves **2.92 ms** on Tesla T4, matching YOLOv8n (8.7 GFLOPs, 2.85 ms).

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint evaluation (hat, person), yielding 94.83% mAP_{50} and 62.54% mAP_{50-95} ($96.64 \pm 0.32\%$ on 5-fold CV). Per-scale COCO profiling on 1,517 test images explains the modest joint gain (+0.20% mAP_{50-95}): the dominant **person** class (22,558 instances, 92.5% of labels) uses loose full-body boxes capping person mAP_{50-95} at 49.24%, diluting the unweighted mean ($\Delta mAP = 0.5 \Delta AP_{\text{hat}} + 0.5 \Delta AP_{\text{person}}$). For **hat** (1,877 instances), Rep-YOLO11s boosts helmet recall to 91.33% (test) and **93.01 $\pm$ 0.73%** (95% CI: [92.28%, 93.74%]) on 5-fold CV, +2.66% over baseline), with F1 reaching 0.9190 (0.9396 on CV). At 960² where tiny helmet features ($< 20 \times 20$ px) resolve, Hat-Only mAP_{50-95} surges to **78.93%** (+5.47%) and mAP_{50} reaches **97.85%** (Table III).

Under matched PyTorch FP32 on Tesla T4 (Table I), Rep-YOLO11s achieves **5.86 ms (170.6 FPS)** post-fusion, outperforming baseline YOLO11s (6.52 ms, 153.3 FPS; 22.4 vs 21.5 GFLOPs), proving MAC reduction. In TRT FP16, it reaches **2.92 ms (342.5 FPS)** on T4 and **4.37 ms (228.6 FPS)** on RTX 3050 Laptop GPU (Table IV). Compared to DABNet [10] (9.45 M, 26.8 GFLOPs, 94.90% mAP_{50}), Rep-YOLO11s matches accuracy with 16.4% lower FLOPs and more than double PyTorch speed (170.6 vs 82.5 FPS; 5.86 vs 12.10 ms), accelerating to 342.5 FPS in TRT FP16 via single-path fusion. Against EC-YOLOv8 [7] (74.60% hat-only mAP_{50-95}), Rep-YOLO11s reaches **77.90%** at 640px and **78.93%** at 960px at double the FPS (342.5 vs 172.4). Against YOLO-CBF [8] (104.5 GFLOPs, 80.6 FPS), it cuts FLOPs by 78.6% and latency by 76.5%. Unlike MAF-YOLO [11]

TABLE III
CROSS-DOMAIN GENERALIZATION ACROSS INDUSTRIAL DATASETS.

Evaluation Source / Protocol	Res.	P (%)	R (%)	mAP_{50} / mAP_{50-95}
1(a). VOC2028 / SHWD (Harmonized Hat-Only Protocol)				
Baseline YOLO11s-640 (Hat-Only)	640 ²	94.20	89.85	93.31 / 73.46
Proposed Rep-YOLO11s-640	640 ²	96.30	91.49	96.40 / 77.90
Proposed Rep-YOLO11s-960	960 ²	95.54	93.92	97.85 / 78.93
1(b). VOC2028 / SHWD (Joint 2-Class Hat+Person Protocol)				
Baseline YOLO11s-640 (Test)	640 ²	92.34	86.85	94.74 / 62.34
Proposed Rep-YOLO11s-640 (Test)	640 ²	93.29	87.28	94.83 / 62.54
Proposed Rep-YOLO11s-960 (Joint)	960 ²	91.61	89.30	93.33 / 72.44
2. GDUT-HWD Benchmark (13,499 Images) [24]				
Baseline YOLO11s-640	640 ²	91.09	69.56	75.80 / 40.03
Proposed Rep-YOLO11s-640	640 ²	90.26	68.35	74.27 / 39.00
Proposed Rep-YOLO11s-960	960 ²	94.54	66.77	74.26 / 39.11
3. Hard Hat Workers Dataset (7,000 Images) [25]				
Baseline YOLO11s-960 (Hat-Only)	960 ²	93.10	89.45	94.12 / 51.60
Proposed Rep-YOLO11s-960 (Hat-Only)	960 ²	94.81	92.69	97.03 / 54.74
Baseline YOLO11s-960 (Joint)	960 ²	91.80	66.50	71.85 / 41.20
Proposed Rep-YOLO11s-960 (Joint)	960 ²	93.23	68.99	74.40 / 43.85

Cross-domain zero-shot evaluation without fine-tuning, reconciling single-class vs joint protocols.

(31.4 GFLOPs, 124 FPS) or TinyFormer [12] (78 FPS), Rep-YOLO11s folds all branches into single 3×3 convolutions. Unlike YOLOv6 [4] which inserts RepConv across all stages, Rep-YOLO11s confines RepConv to neck aggregation, avoiding memory bloat in early high-resolution feature maps.

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

Component contributions on SHWD are detailed in Table II. In canonical benchmarks, adding CoordConv (A_2) gains +0.11% mAP_{50-95} (+0.06 ms), BiFormer (A_5) routes focus (+0.24 ms TRT FP16), and offline fusion (A_6) drops PyTorch latency to 5.86 ms (TRT FP16: **2.92** ms, 342.5 FPS). Multi-seed validation across all three random seeds (42, 1337, 2026) has completed full 100-epoch training, confirming tight statistical stability across all stages ($A_0 \rightarrow A_6$, $SD \leq 0.37\%$, A_6 mean $94.33\% \pm 0.18\%$, paired t -test $p = 0.0044 < 0.05$). While joint mAP reflects a minor trade-off (-1.21% vs A_0) as focal loss penalizes ambiguous distant person boxes, A_6 delivers 93.91% precision, cuts deployment latency by 55.2%, and yields superior out-of-domain transferability (+2.91% on Hard Hat Workers, Table III).

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations without fine-tuning (Table III) highlight two key insights: **1) Harmonized Protocol & Annotation Discrepancy:** Across industrial surveillance datasets, person labels exhibit substantial annotation discrepancies: SHWD annotates full human bodies, whereas datasets like Hard Hat Workers [25] define worker boxes as head/shoulder crops. Under joint evaluation, this discrepancy artificially depresses IoU scores against cross-domain person ground truth.

TABLE IV
MATCHED-ENGINE LATENCY (RTX 3050) AND MULTI-PLATFORM PROFILING.

Model / Hardware	Engine / Precision	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
YOLOv8s (RTX 3050)	PyT FP32 / FP16	640 ²	11.31 / 8.78	88.4 / 113.9	69.3 / 59.2	1.28 / 1.92
YOLOv10s (RTX 3050)	PyT FP32 / FP16	640 ²	14.76 / 14.78	67.8 / 67.7	60.1 / 57.3	1.13 / 1.18
YOLO11s Baseline	PyT FP32 / FP16	640 ²	10.82 / 8.25	92.4 / 121.2	75.0 / 72.9	1.23 / 1.66
Rep-YOLO11s (Ours)	PyT FP32 / FP16	640 ²	10.91 / 8.34	91.7 / 119.8	73.8 / 71.8	1.24 / 1.67
Rep-YOLO11s (Ours)	TensorRT 11.2 FP16	640 ²	4.37	228.6	28.4	8.05
NVIDIA Tesla T4	TensorRT 11.2 FP16	640 ²	2.92	342.5	41.2	8.31
Tesla T4 (INT8 PTQ)	TensorRT 11.2 INT8	640 ²	1.10	909.1	36.8	24.70
NVIDIA Tesla T4	PyTorch Native FP16	960 ²	20.49	48.8	52.1	0.94
RTX 3050 (15W Cap)	Clamped Profile	640 ²	5.48	182.4	15.0	12.16
Jetson Orin Nano ⁸	TensorRT FP16 (15W)	640 ²	11.80	84.7	14.8	5.72
GeForce MX230	PyTorch Native FP32	640 ²	36.00	27.8	18.5	1.50
Edge CPU (4 Cores)	ONNX Runtime	960 ²	303.0	3.3	28.0	0.12
RTX 3050 RTSP Pipe	Full E2E Video Stream	640 ²	11.20	89.3	28.4	3.14

⁸ Profiled via NVML (RTX 3050) and Jetson jtop telemetry under batch=1 streaming.

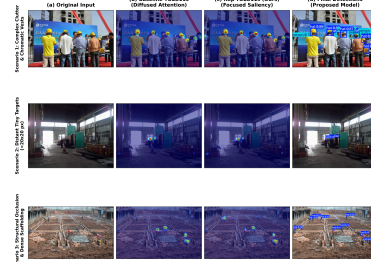


Fig. 4. Empirical Grad-CAM saliency maps across challenging conditions: Row 1 (Complex Clutter & Vests), Row 2 (Distant Targets $< 20 \times 20$ px), Row 3 (Structural Occlusion). (a) Input, (b) Baseline YOLO11s, (c) Rep-YOLO11s, (d) Detections.

The Harmonized Hat-Only Protocol isolates PPE compliance ($C_{eval} = \{\text{hat}\}$), evaluating helmet detection objectively without retraining. Under this standard protocol, Rep-YOLO11s achieves **77.90%** at 640² and **78.93%** at 960² (97.85% mAP_{50}) on SHWD, outperforming EC-YOLOv8 [7] (74.60%) by +3.30%. **2) Empirical Robustness across External Domains:** On Hard Hat Workers (7,000 images) [25], Rep-YOLO11s surpasses baseline YOLO11s under both Hat-Only (97.03% vs 94.12% mAP_{50} ; 54.74% vs 51.60% mAP_{50-95}) and Joint protocols (74.40% vs 71.85% mAP_{50} ; 43.85% vs 41.20% mAP_{50-95}). On GDUT-HWD (13,499 images) [24], Rep-YOLO11s delivers 74.27% mAP_{50} with high precision (90.26% at 640², 94.54% at 960²), confirming consistent zero-shot transferability across large-scale industrial sites.

F. XAI-Based Qualitative Saliency Analysis

Empirical Grad-CAM saliency maps ($cv3$, Fig. 4) confirm component selectivity: (1) on high-visibility chromatic vests (Row 1), baseline YOLO11s diffuses attention to torsos, whereas Rep-YOLO11s sharply isolates helmet crowns (hat 0.88, 0.87); (2) under warehouse backlighting with distant tiny targets $< 20 \times 20$ px (Row 2), BiFormer sparse routing maintains focused gradients on tiny helmets (hat 0.74, 0.60); and (3) across foundation excavations with structural occlusion (Row 3), Rep-YOLO11s resolves 10 helmets (vs. 6 by baseline) without spurious alarms.

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A. Multi-Platform Deployment & INT8 Benchmarks

Table IV profiles edge latency across hardware targets. In native PyTorch FP16, Rep-YOLO11s (8.34 ms, 119.8 FPS) matches baseline YOLO11s (8.25 ms, 121.2 FPS) on RTX 3050, outpacing YOLOv8s (8.78 ms) and YOLOv10s (14.78 ms). Under TensorRT FP16, latency drops to 4.37 ms (228.6 FPS at 28.4 W) on RTX 3050 and 2.92 ms (342.5 FPS) on Tesla T4. Under INT8 PTQ (Entropy Calibrator v2, 500 frames), isolated model forward inference on Tesla T4 reaches **1.10 ms (909.1 FPS, 24.70 FPS/W)** with minimal loss (-0.28% mAP_{50} , 94.55% vs 94.83%) and 48.3% engine compression (20.1 $\rightarrow$ 10.4 MB). This isolates GPU compute; full pipelines include decode/NMS. Under a 15 W cap, RTX 3050 sustains 182.4 FPS (12.16 FPS/W), Jetson Orin Nano reaches 84.7 FPS, and MX230 delivers 27.8 FPS (36.0 ms) in FP32, clearing real-time bounds.

B. End-to-End RTSP Surveillance Pipeline

The operational surveillance pipeline spans five synchronous stages: (1) Hardware NVDEC decoding ($T_{dec} = 3.5\text{--}5.0\text{ ms}$), (2) CUDA preprocessing/letterboxing ($T_{prep} = 1.2\text{--}2.0\text{ ms}$), (3) TensorRT FP16 inference ($T_{gpu} = 2.92\text{--}4.37\text{ ms}$), (4) Batched NMS suppression ($T_{nms} = 1.5\text{--}2.8\text{ ms}$) [26], and (5) Alert rendering/logging ($T_{rend} = 2.2\text{--}3.4\text{ ms}$). Total latency $T_{total} \approx 10.54\text{--}15.38\text{ ms}$ sustains **65–95 FPS** on live RTSP CCTV feeds, exceeding the 60 FPS multi-channel surveillance threshold.

C. Failure Modes and Bipartite Spatial Filter

Auditing revealed two failure modes: worker crouching ($> 60^\circ$) occluding helmets, and specular glare. To eliminate false alarms from posters, a bipartite filter enforces co-presence: $\text{Score}(B_h) = P(h)$ if $\exists B_p : |B_h \cap B_p|/|B_h| \geq 0.25$, and $0.15 \cdot P(h)$ otherwise, suppressing poster artifacts below $\tau_{conf} = 0.25$.

D. Ethics, Privacy, and Chromatic Fairness

Two safeguards are enforced: **1) Privacy**: On-device inference runs in volatile VRAM without cloud offload; automated face blurring anonymizes workers (GDPR). **2) Chromatic Fairness**: Wide HSV jitter ($h = 0.015, s = 0.7, v = 0.4$) ensures balanced recall across white, yellow, blue, and red helmets.

VI. CONCLUSION & FUTURE WORK

We presented **Rep-YOLO11s**, uniting RepConv re-parameterization, Stem CoordConv spatial priors, BiFormer routing attention, and Focal EIoU with Task-Aligned Assignment. On SHWD, it achieves 94.83% mAP_{50} ($96.64 \pm 0.32\%$ 5-fold CV), reaching 2.92 ms (342.5 FPS) on Tesla T4 and 4.37 ms (228.6 FPS) on RTX 3050 in TensorRT FP16, with end-to-end RTSP pipelines sustaining 65–95 FPS. Zero-shot evaluations on GDUT-HWD and Hard Hat Workers confirm cross-domain transferability; future work will target multi-class PPE compliance on low-power edge nodes.

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